

The Ising Model

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Friedli & Velenik, *Statistical Mechanics of Lattice Systems*

We study the Ising model on $\mathbb{Z}^d$, giving precise definitions and a detailed analysis of the conditions ensuring the existence or absence of a phase transition. The plan: §3.1 defines the model with various boundary conditions; §3.2 introduces the thermodynamic limit, the pressure and the magnetization, computed explicitly for $d = 1$ in §3.3; §3.4 defines infinite-volume Gibbs states; §3.6 develops correlation inequalities (GKS, FKG); §3.7 analyzes the phase diagram, proving a phase transition at $h = 0$, $d \geq 2$, β large (Peierls), uniqueness at high temperature, and uniqueness at $h \neq 0$ (Lee–Yang); §3.10 collects complements.

3.1 Finite-volume Gibbs distributions

Configurations of the Ising model in a finite volume $\Lambda \Subset \mathbb{Z}^d$ with *free* boundary condition form $\Omega_\Lambda = \{-1, 1\}^\Lambda$. The spin at i is $\sigma_i(\omega) = \omega_i$. With $\mathcal{E}_\Lambda = \{\{i, j\} \subset \Lambda : i \sim j\}$ the nearest-neighbour edges, the energy of ω is

$$\mathcal{H}_{\Lambda; \beta, h}^\varnothing(\omega) = -\beta \sum_{\{i, j\} \in \mathcal{E}_\Lambda} \sigma_i(\omega) \sigma_j(\omega) - h \sum_{i \in \Lambda} \sigma_i(\omega),$$

with inverse temperature $\beta \in \mathbb{R}_{\geq 0}$ and magnetic field $h \in \mathbb{R}$. The superscript $\varnothing$ marks the free boundary condition.

Definition 3.1. The Gibbs distribution of the Ising model in Λ with free boundary condition is the probability measure on Ω_Λ

$$\mu_{\Lambda; \beta, h}^\varnothing(\omega) = \frac{e^{-\mathcal{H}_{\Lambda; \beta, h}^\varnothing(\omega)}}{Z_{\Lambda; \beta, h}^\varnothing}, \quad Z_{\Lambda; \beta, h}^\varnothing = \sum_{\omega \in \Omega_\Lambda} e^{-\mathcal{H}_{\Lambda; \beta, h}^\varnothing(\omega)}.$$

Definition 3.2. The Gibbs distribution in the box V_n with *periodic* boundary condition is defined analogously, with the Hamiltonian summed over the edges of the torus (nearest-neighbour pairs taken modulo the side length).

Definition 3.3. The Gibbs distribution of the Ising model in Λ with boundary condition $\eta \in \Omega$ uses the Hamiltonian

$$\mathcal{H}_{\Lambda; \beta, h}^\eta(\omega) = -\beta \sum_{\{i, j\} \cap \Lambda \neq \varnothing} \sigma_i \sigma_j - h \sum_{i \in \Lambda} \sigma_i,$$

where spins outside Λ are frozen to η . The special cases $\eta \equiv +1$ and $\eta \equiv -1$ give the $+$ and $-$ boundary conditions.

3.2 Thermodynamic limit, pressure and magnetization

3.2.1 Convergence of subsets

We write $\Lambda \uparrow \mathbb{Z}^d$ (van Hove convergence) for an increasing sequence of finite volumes invading $\mathbb{Z}^d$ with vanishing relative boundary. The boxes $B(n) = \{-n, \dots, n\}^d$ are the canonical example.

3.2.2 Pressure

Definition 3.4. The (finite-volume) pressure with boundary condition of type $\#$ is

$$\psi_\Lambda^\#(\beta, h) = \frac{1}{|\Lambda|} \log Z_{\Lambda; \beta, h}^\#.$$

Lemma 3.5. For each type of boundary condition $\#$, the map $(\beta, h) \mapsto \psi_\Lambda^\#(\beta, h)$ is convex.

Theorem 3.6. In the thermodynamic limit, the pressure

$$\psi(\beta, h) = \lim_{\Lambda \uparrow \mathbb{Z}^d} \psi_\Lambda^\#(\beta, h)$$

is well defined, independent of the sequence $\Lambda \uparrow \mathbb{Z}^d$ and of the type of boundary condition. Moreover ψ is convex on $\mathbb{R}_{\geq 0} \times \mathbb{R}$ and even in h .

The proof uses subadditivity for the free boundary condition, then shows boundary conditions and the choice of sequence do not affect the limit.

3.2.3 Magnetization

The average magnetization density in Λ is $m_\Lambda^\#(\beta, h) = \frac{1}{|\Lambda|} \sum_{i \in \Lambda} \langle \sigma_i \rangle_\Lambda^\# = \partial_h \psi_\Lambda^\#$. Let $\mathcal{B}_\beta$ be the (at most countable) set of h at which $h \mapsto \psi(\beta, h)$ fails to be differentiable.

Corollary 3.7. For all $h \notin \mathcal{B}_\beta$, the average magnetization density $m(\beta, h) = \partial_h \psi(\beta, h)$ exists and equals $\lim_{\Lambda \uparrow \mathbb{Z}^d} m_\Lambda^\#(\beta, h)$, independently of the boundary condition.

3.2.4 A first definition of phase transition

Definition 3.8. The pressure ψ exhibits a *first-order phase transition* at (β, h) if $h \mapsto \psi(\beta, h)$ is not differentiable there, i.e. $\frac{\partial \psi}{\partial h^-}(\beta, h) < \frac{\partial \psi}{\partial h^+}(\beta, h)$.

3.3 The one-dimensional Ising model

Theorem 3.9 ($d = 1$). For all $\beta \geq 0$ and $h \in \mathbb{R}$, the pressure of the one-dimensional Ising model is

$$\psi(\beta, h) = \log \left\{ e^\beta \cosh(h) + \sqrt{e^{2\beta} \cosh^2(h) - 2 \sinh(2\beta)} \right\}. \quad (3.10)$$

In particular $h \mapsto \psi(\beta, h)$ is real-analytic for every $\beta \geq 0$, so $\mathcal{B}_\beta = \emptyset$: there is no phase transition in $d = 1$.

The proof uses the transfer-matrix method: Z is a trace of a product of 2×2 matrices, and ψ is the log of the largest eigenvalue.

Theorem 3.10 ($d = 1$). Let $0 < \beta < \infty$ and $\Lambda_n \uparrow \mathbb{Z}$ with arbitrary boundary condition. For all $\epsilon > 0$ there is $c = c(\beta, \epsilon) > 0$ such that, for n large,

$$\mu_{\Lambda_n; \beta, 0}^\#(m_{\Lambda_n} \notin (-\epsilon, \epsilon)) \leq e^{-c|\Lambda_n|}. \quad (3.12)$$

(A large-deviation/concentration statement: at $h = 0$ the empirical magnetization concentrates at 0.)

3.4 Infinite-volume Gibbs states

Definition 3.11. A function $f : \Omega \rightarrow \mathbb{R}$ is *local* if it depends on finitely many spins, i.e. there is $\Delta \Subset \mathbb{Z}^d$ with $f(\omega) = f(\omega')$ whenever $\omega|_\Delta = \omega'|_\Delta$.

Definition 3.13. An (infinite-volume) *state* is a normalized, positive linear functional $\langle \cdot \rangle$ on local functions.

Definition 3.14. Given $\Lambda_n \uparrow \mathbb{Z}^d$ and boundary conditions $(\#_n)$, the associated infinite-volume Gibbs state is the limit $\langle f \rangle = \lim_n \langle f \rangle_{\Lambda_n; \beta, h}^{\#_n}$ for all local f , when it exists.

Definition 3.16. A state $\langle \cdot \rangle$ is *translation invariant* if $\langle f \circ \theta_j \rangle = \langle f \rangle$ for every local f and every $j \in \mathbb{Z}^d$.

Theorem 3.17. Let $\beta \geq 0$, $h \in \mathbb{R}$. Along any sequence $\Lambda_n \uparrow \mathbb{Z}^d$, the finite-volume Gibbs distributions with $+$ (resp. $-$) boundary condition converge to infinite-volume Gibbs states

$$\langle \cdot \rangle_{\beta, h}^+ = \lim_n \langle \cdot \rangle_{\Lambda_n; \beta, h}^+, \quad \langle \cdot \rangle_{\beta, h}^- = \lim_n \langle \cdot \rangle_{\Lambda_n; \beta, h}^-,$$

independently of the sequence. These are translation-invariant Gibbs states, extremal in the sense that $\langle \cdot \rangle^+ \geq \langle \cdot \rangle \geq \langle \cdot \rangle^-$ on nondecreasing local functions.

The proof relies on monotonicity in the volume and boundary condition (FKG / Lemma 3.22–3.23) together with the polynomial expansion Lemma 3.19.

3.5 Two families of local functions

Lemma 3.19. Every local f admits a finite expansion $f = \sum_{A \subset \text{supp}(f)} \hat{f}_A \sigma_A$ in the monomials $\sigma_A = \prod_{i \in A} \sigma_i$, with real coefficients $\hat{f}_A$. Consequently agreement of all correlation functions $\langle \sigma_A \rangle$ determines a state.

The two basic monotone families are σ_A and the occupation variables $n_i = \frac{1+\sigma_i}{2}$, $n_A = \prod_{i \in A} n_i$.

3.6 Correlation inequalities

3.6.1 The GKS inequalities

Theorem 3.20 (GKS inequalities). Let $J \geq 0$, $h \geq 0$ and $\Lambda \Subset \mathbb{Z}^d$. Then for all $A, B \subset \Lambda$,

$$\langle \sigma_A \rangle_{\Lambda; J, h}^+ \geq 0, \quad \langle \sigma_A \sigma_B \rangle_{\Lambda; J, h}^+ \geq \langle \sigma_A \rangle_{\Lambda; J, h}^+ \langle \sigma_B \rangle_{\Lambda; J, h}^+. \quad (3.21\text{--}3.22)$$

The inequalities also hold for free and periodic boundary conditions.

3.6.2 The FKG inequality

Theorem 3.21 (FKG inequality). *Let $J = (J_{ij}) \geq 0$, $h = (h_i) \in \mathbb{R}^{\mathbb{Z}^d}$ arbitrary, $\Lambda \Subset \mathbb{Z}^d$, and $\#$ any boundary condition. Then for any pair of nondecreasing functions f, g ,*

$$\langle fg \rangle_{\Lambda; J, h}^{\#} \geq \langle f \rangle_{\Lambda; J, h}^{\#} \langle g \rangle_{\Lambda; J, h}^{\#}. \quad (3.24)$$

(The inequality reverses if g is nonincreasing.)

3.6.3 Consequences (monotonicity)

Lemma 3.22. *For nondecreasing f and $\Lambda_1 \subset \Lambda_2 \Subset \mathbb{Z}^d$, $\langle f \rangle_{\Lambda_1}^+ \geq \langle f \rangle_{\Lambda_2}^+$ and $\langle f \rangle_{\Lambda_1}^- \leq \langle f \rangle_{\Lambda_2}^-$: increasing the volume under $+$ b.c. decreases the expectation of any nondecreasing f (and conversely under $-$).*

Lemma 3.23. *For any nondecreasing f , $\beta \geq 0$ and arbitrary boundary condition η , $\langle f \rangle_{\Lambda; \beta, h}^- \leq \langle f \rangle_{\Lambda; \beta, h}^{\eta} \leq \langle f \rangle_{\Lambda; \beta, h}^+$. The $+$ and $-$ states dominate all others. These monotonicity facts are what make the limits in Theorem 3.17 exist.*

3.7 Phase diagram

Definition 3.24. If at least two distinct Gibbs states can be constructed for a pair (β, h) , one says there is *non-uniqueness* (a phase transition) at (β, h) .

Theorem 3.25 (main result).

1. In any $d \geq 1$, when $h \neq 0$ there is a unique Gibbs state for all $\beta \in \mathbb{R}_{\geq 0}$.
2. In $d = 1$, there is a unique Gibbs state at every (β, h) .
3. When $h = 0$ and $d \geq 2$, there exists $\beta_c = \beta_c(d) \in (0, \infty)$ such that:
 - for $\beta < \beta_c$ the Gibbs state at $(\beta, 0)$ is unique;
 - for $\beta > \beta_c$ a first-order phase transition occurs, $\langle \cdot \rangle_{\beta, 0}^+ \neq \langle \cdot \rangle_{\beta, 0}^-$ (spontaneous magnetization $m^*(\beta) > 0$).

The proof is spread over the next sections: item 3 (existence of a transition) by the Peierls argument and high-temperature uniqueness; items 1–2 by the Lee–Yang analyticity argument.

3.7.1 Two criteria for (non)-uniqueness

Theorem 3.28. *For $(\beta, h) \in \mathbb{R}_{\geq 0} \times \mathbb{R}$ the following are equivalent:*

1. *there is a unique Gibbs state at (β, h) ;*
2. $\langle \cdot \rangle_{\beta, h}^+ = \langle \cdot \rangle_{\beta, h}^-$;
3. $\langle \sigma_0 \rangle_{\beta, h}^+ = \langle \sigma_0 \rangle_{\beta, h}^-$.

Thus uniqueness is detected by a single one-point function.

Proposition 3.29. *For any $\Lambda \uparrow \mathbb{Z}^d$ the limits $m^+(\beta, h) = \langle \sigma_0 \rangle_{\beta, h}^+$ and $m^-(\beta, h) = \langle \sigma_0 \rangle_{\beta, h}^-$ exist; m^+ is right-continuous, m^- left-continuous in h , and $m^- = -m^+$ by spin-flip symmetry.*

Lemma 3.31. *For all $\beta \geq 0$, $h \mapsto \langle \sigma_0 \rangle_{\beta, h}^+$ is nondecreasing and right-continuous (analogous statements for $\langle \sigma_0 \rangle^-$). This yields the one-sided continuity of $m^\pm$.*

Definition 3.32. The critical inverse temperature is

$$\beta_c(d) = \inf\{\beta \geq 0 : m^*(\beta) > 0\} = \sup\{\beta \geq 0 : m^*(\beta) = 0\},$$

where $m^*(\beta) = \langle \sigma_0 \rangle_{\beta, 0}^+$ is the spontaneous magnetization. At $h = 0$, uniqueness is equivalent to $m^*(\beta) = 0$.

Theorem 3.34. *For all $\beta \geq 0$, $h \in \mathbb{R}$,*

$$\frac{\partial \psi}{\partial h^+}(\beta, h) = m^+(\beta, h), \quad \frac{\partial \psi}{\partial h^-}(\beta, h) = m^-(\beta, h).$$

Hence $h \mapsto \psi(\beta, h)$ is differentiable at h iff there is a unique Gibbs state at (β, h) : the analytic notion of first-order transition (Definition 3.8) coincides with the probabilistic notion of non-uniqueness.

3.7.2 Spontaneous symmetry breaking at low temperature (Peierls)

For $h = 0$, $d = 2$, in the box $B(n)$ with $+$ boundary condition, each configuration is described by a family of *contours* (closed dual circuits separating $+$ and $-$ regions).

Lemma 3.37 (Peierls' estimate). *For all $\beta > 0$ and any contour γ^* ,*

$$\mu_{B(n); \beta, 0}^+(\Gamma \ni \gamma^*) \leq e^{-2\beta|\gamma^*|}. \quad (3.34)$$

The probability of a fixed contour decays exponentially in its length: the ground state η^+ is stable at large β .

Summing over contours surrounding the origin (and controlling their number, Lemma 3.38), one gets $\langle \sigma_0 \rangle_{\beta, 0}^+ > 0$ for β large enough, hence $\beta_c(d) < \infty$ for $d \geq 2$, with the explicit bound $\beta_c(2) < 0.88$ (Exercise 3.21).

Lemma 3.38. *Combinatorial bound on the number of contours of a given length through a fixed edge, obtained by an Eulerian traversal of a connected graph: a connected graph with N edges admits a closed walk using each edge at most twice.*

3.7.3 Uniqueness at high temperature

A high-temperature (small- β) expansion represents the two-point function $\langle \sigma_i \sigma_j \rangle_{\beta, 0}^+$ as a sum over collections of edges/paths. For β small the geometric weight is summable, giving exponential decay of correlations and therefore $m^*(\beta) = 0$, i.e. uniqueness for $\beta < \beta_c$, and $\beta_c(d) > 0$.

3.7.4 Uniqueness in nonzero magnetic field

Theorem 3.40. *Let $\beta > 0$. As a function of h , the thermodynamic-limit pressure $\psi = \psi(h)$ extends from $\{h > 0\}$ (resp. $\{h < 0\}$) to an analytic function on the right half-plane H^+ (resp. H^-); on $H^\pm$ it can be computed via the free-boundary thermodynamic limit.*

By Theorem 3.34 analyticity in h forces a unique Gibbs state, so Theorem 3.40 yields item 1 of Theorem 3.25 (uniqueness for all $h \neq 0$).

Theorem 3.42 (Lee–Yang). *Let $\beta \geq 0$ and let $D \subset \mathbb{C}$ be open, simply connected, with $D \cap \mathbb{R}$ an interval. If, for every finite volume $\Lambda \Subset \mathbb{Z}^d$, the partition function satisfies $Z_{\Lambda;\beta,h}^\emptyset \neq 0$ for all $h \in D$, then the pressure $h \mapsto \psi(h)$ admits an analytic continuation to D .*

Theorem 3.43 (Lee–Yang Circle Theorem). *The hypothesis of Theorem 3.42 holds with $D = H^+$ and $D = H^-$. Equivalently, in the variable $z = e^{-2h}$ all zeros of the partition function lie on the unit circle $|z| = 1$ (i.e. at purely imaginary h). This is what supplies the no-zero condition needed for Theorem 3.40.*

3.7.5 Summary of what has been proved

Combining: a first-order phase transition at $(\beta, 0)$ for $\beta > \beta_c(d)$, $d \geq 2$ (Peierls); uniqueness for $\beta < \beta_c(d)$ (high-temperature expansion); and uniqueness for all $h \neq 0$ and all d (Lee–Yang). Together these establish the full phase diagram of Theorem 3.25.

3.8 Proofs of the correlation inequalities

§3.8.1 proves the GKS inequalities (Theorem 3.20) by expanding products of σ_A and using that all coefficients of $e^{\beta\sigma_i\sigma_j}$, $e^{h\sigma_i}$ are nonnegative when $\beta, h \geq 0$. §3.8.2 proves the FKG inequality (Theorem 3.21) from the FKG lattice condition (Holley’s inequality / the Harris–FKG correlation inequality for monotone measures).

3.9 Bibliographical references

Pointers to Ising, Peierls, Griffiths–Kelly–Sherman, Fortuin–Kasteleyn–Ginibre, Lee–Yang, and the historical literature on the model.

3.10 Complements

A series of more advanced or specialized topics, treated less formally:

- 3.10.1 **Kramers–Wannier duality** — the self-duality of the 2D model relating high- and low-temperature regimes and locating the self-dual point β_{sd} , conjecturally $= \beta_c(2)$.
- 3.10.2 **Mean-field bounds** — comparison inequalities bounding m^* above/below by mean-field quantities.
- 3.10.3 **An alternative proof of the FKG inequality.**
- 3.10.4 **Transfer matrix and Markov chains** — the operator viewpoint and its probabilistic interpretation.
- 3.10.5 **The Ising antiferromagnet** ($J < 0$) and the role of bipartiteness.
- 3.10.6 **Random-cluster and random-current representations** — the FK/current geometric reformulations, percolation of clusters, and the switching lemma.
- 3.10.7 **Non-translation-invariant Gibbs states and interfaces** — Dobrushin states in $d \geq 3$, rigid interfaces, and their absence in $d = 2$.

- 3.10.8 **Gibbs states and local behavior in large finite systems.**
- 3.10.9 **Absence of analytic continuation of the pressure** through the transition point.
- 3.10.10 **Metastable behavior in finite systems.**
- 3.10.11 **Critical phenomena** — critical exponents, scaling, and what is/ is not rigorously known.
- 3.10.12 **Exact solution** — Onsager's solution in $d = 2$ and the explicit $\beta_c(2) = \frac{1}{2} \log(1 + \sqrt{2})$.
- 3.10.13 **Stochastic dynamics** — Glauber dynamics reversible w.r.t. the Gibbs measure.